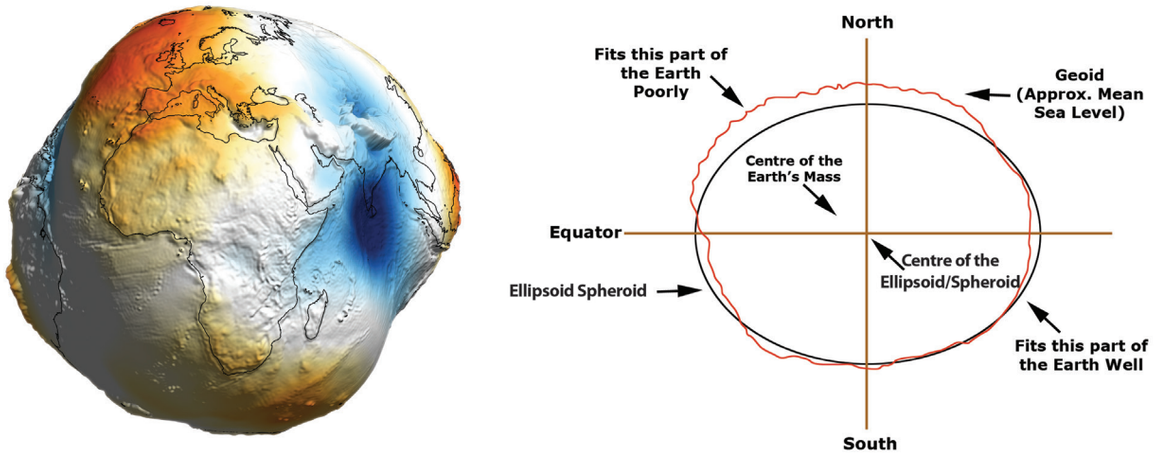


Figure 2.9 Geoid : Shape of the earth

(275-195 BCE) estimated size of earth from observations that the elevation of the sun varied with position on the Earth's surface in Egypt.The Following Observations Show that the earth is sphere in shape

1. Mountain peaks lit by the Sun after sunset.
2. Ships disappear below the horizon as they sail across ocean.
3. The moon looks like a disc.
4. The Earth casts a circular shadow during lunar eclipses.

The Earth is an oblate spheroid, bulged at the equator and flattened at the poles. It is called 'Geoid' (Figure 2.9) meaning the earth is earth-shaped. The bulge at the equator is caused by the centrifugal force of the Earth's rotation. The gravitational pull of the earth is the strongest at the flattened poles and it is weaker towards the equator.

HOTS

Chimborazo in Ecuador is higher than Mount Everest, if measured from the centre of the Earth. Why?

The Sun's gravitational pull differs in force at the poles. The North Pole points in the same direction to the North Star when it revolves about the Sun. If the Earth would not have been tilted on its axis, the days and nights would have been of same duration always.

2.14 Motions of the earth

The earth has two basic movements: 1) Rotation and 2) Revolution.



Galactic movement: This is the movement of the earth with the sun and the rest of the solar system in an orbit around the centre of the MilkyWay Galaxy. This, however, has little effect upon the changing environment of the earth.

1. **Rotation:** The spinning of the earth around its axis is called the rotation of the earth. The axis is the imaginary line passing through the centre of the earth. The earth completes one rotation in 23 hours, 56 minutes and 4.09 seconds. It rotates in an eastward direction opposite

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